

TRAIN WITH
SHUBHAM

JIRA

Project Management Tool
Integration With **Github**



Introduction:

What is Jira?

Jira is like the central command center for your projects. It keeps everything organized and helps your team communicate effectively. Whether you're a small team working on a simple app or a large company managing multiple projects, Jira can scale to meet your needs.

Basically, its a software developed by a company called Atlassian. It's a powerful tool used mainly for three things:

- 1.Project Management
- 2.Issue Tracking
- 3.Agile Software Development

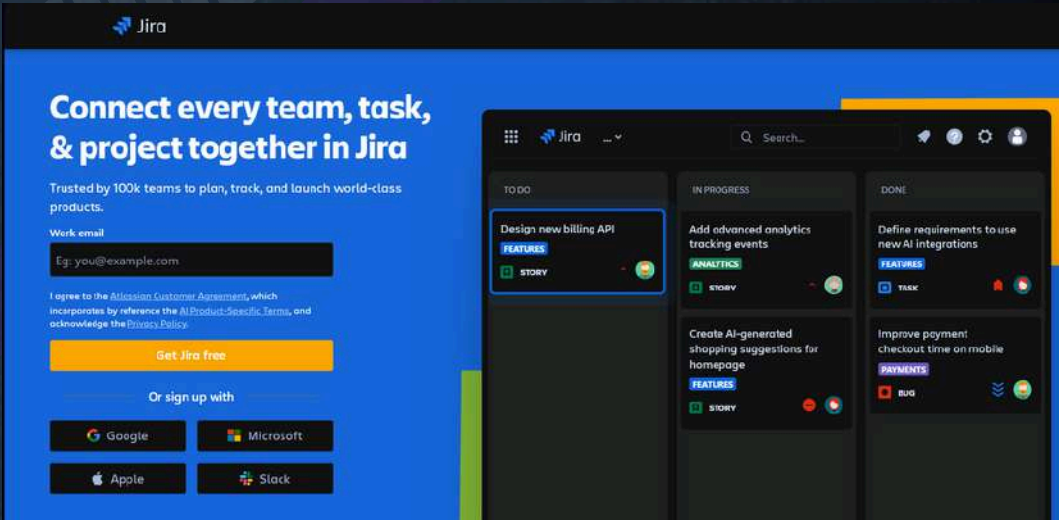
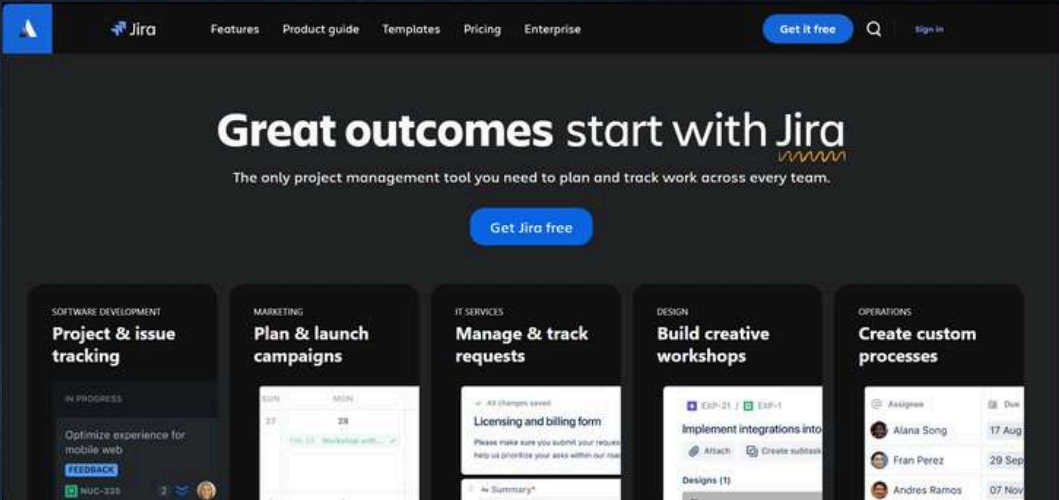
Key Features of Jira

Customizable Workflows: Workflows in Jira define how tasks move from start to finish. You can customize these workflows to fit your team's needs. For example, a workflow for fixing a bug might look like this:

- Story
- Story Point
- Epic
- To Do
- In Progress
- In Review
- Done
- Sprint
- Velocity Chart

Detailed Reporting: Jira provides various reports to help you understand how your project is progressing. For example, you can see how many tasks are completed, how many are still in progress, and how many are yet to be started. This helps in planning and improving future projects.

Robust Integration Options: Jira can connect with many other tools your team might be using. For example, it integrates with GitHub (a code hosting platform) so that code changes can be linked to tasks in Jira. It also integrates with Slack (a communication tool) to get updates about your project directly in your chat.



Story vs Epic

In Jira, both Stories and Epics are used to organize work but they serve different purposes and scopes.

A **Story** represents a **single piece of work** that needs to be **completed within a sprint**. It's usually a **smaller task or requirement** that a team can accomplish in a short time frame. Stories often **follow a user-focused format** like "As a user, I want to be able to log in so that I can access my account."

An **Epic**, on the other hand, is a **larger body of work that can be broken down into multiple stories**. Epics often **span multiple sprints** and are used to **manage more complex projects** or features that require substantial effort and time to complete. For example, an epic might be "Implement user authentication system," which would then be broken down into stories like "Create login page," "Set up database for user credentials," and "Implement password reset feature."

Sprint

- **Definition:** A sprint is a **time-boxed period, typically 1-4 weeks**, during which a specific set of **work must be completed** and made ready for review.
- **Purpose:** Allows teams to **break down their work into manageable chunks** and deliver increments of value regularly.
- **Process:**
 - **Sprint Planning:** Define what can be delivered in the sprint and how that work will be achieved.
 - **Daily Standups:** Short meetings to discuss progress and obstacles.
 - **Sprint Review:** At the end of the sprint, review what was accomplished.
 - **Sprint Retrospective:** Reflect on the sprint to identify improvements.

Velocity Chart

- **Definition:** A velocity chart tracks the **amount of work a team completes during each sprint**.
- **Purpose:** Helps **predict how much work a team can realistically achieve** in future sprints.
- **Components:**
 - **Velocity:** The average amount of story points completed in previous sprints.
 - **Displayed:** As a bar chart showing story points completed per sprint.
- **Usage:** Teams use the velocity to plan sprints by understanding their capacity.

Story Points

- **Definition:** Story points are *units of measure for expressing the overall effort* required to fully implement a product backlog item or any other piece of work.
- **Purpose:** Helps estimate the complexity, effort, and time required to complete a story.
- **Assignment:**
 - **Relative Estimation:** Compare stories to each other to assign points.
 - **Common Scale:** Fibonacci sequence (1, 2, 3, 5, 8, 13, etc.) is often used.
- **Example:** A simple task might be 1 point, a more complex task 5 points, and a very complex task 13 points.

Dashboards and Gadgets

Jira's Dashboards provide a customizable interface to display key project information using various gadgets. Dashboards help in visualizing data and tracking progress at a glance. Gadgets include:

- **Issue Statistics:** Show statistics on issue types, priorities, or custom fields.
- **Sprint Burndown:** Visualize the progress of a sprint.
- **Pie Charts:** Display issue distribution by status, assignee, or other fields.

Permissions and Security

Jira provides robust permissions and security settings to control access and ensure data integrity. Features include:

- **Project Roles:** Define roles and assign permissions based on team members' responsibilities.
- **Issue Security:** Set security levels to restrict access to specific issues.
- **Field-Level Security:** Control visibility and editability of custom fields based on user roles.

By leveraging these additional features and concepts, you can maximize the potential of Jira for your project management and agile development needs, making it a comprehensive tool for tracking, planning, and collaborating effectively.

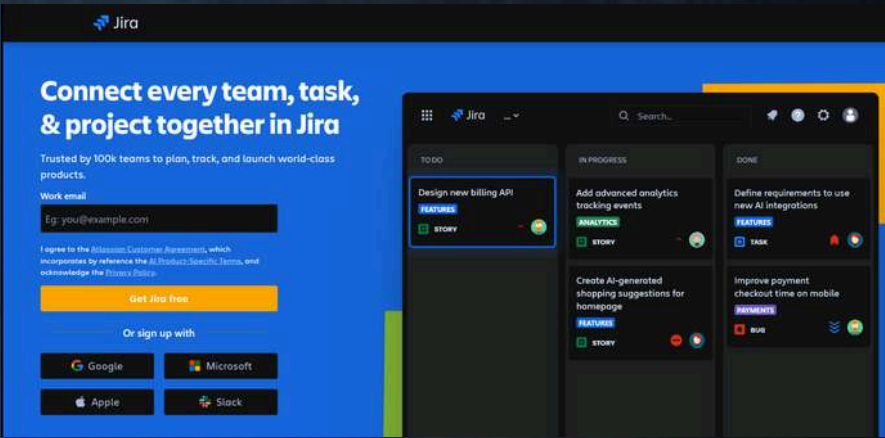
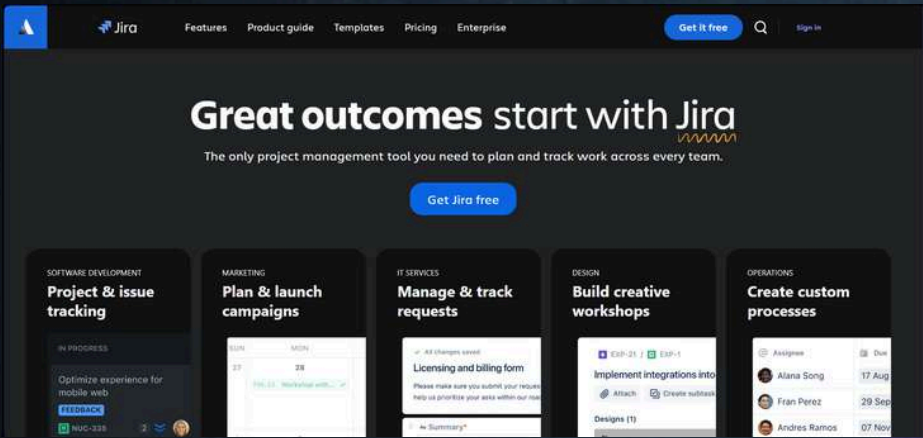
Account Creation:

Creating a Jira account is a straightforward process. Here's a step-by-step guide to help you get started:

1. Sign Up for Jira

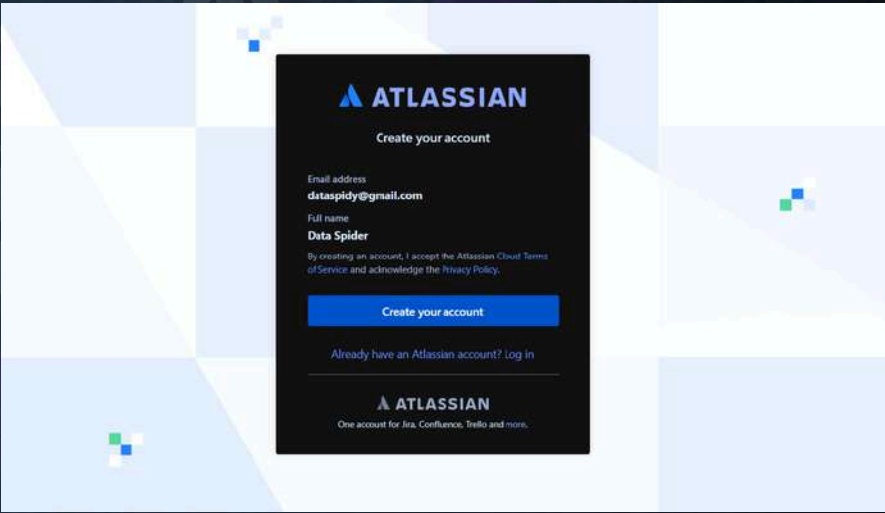
Visit the Jira Website

- Step: Open your web browser and go to the Jira website:
<https://www.atlassian.com/software/jira>



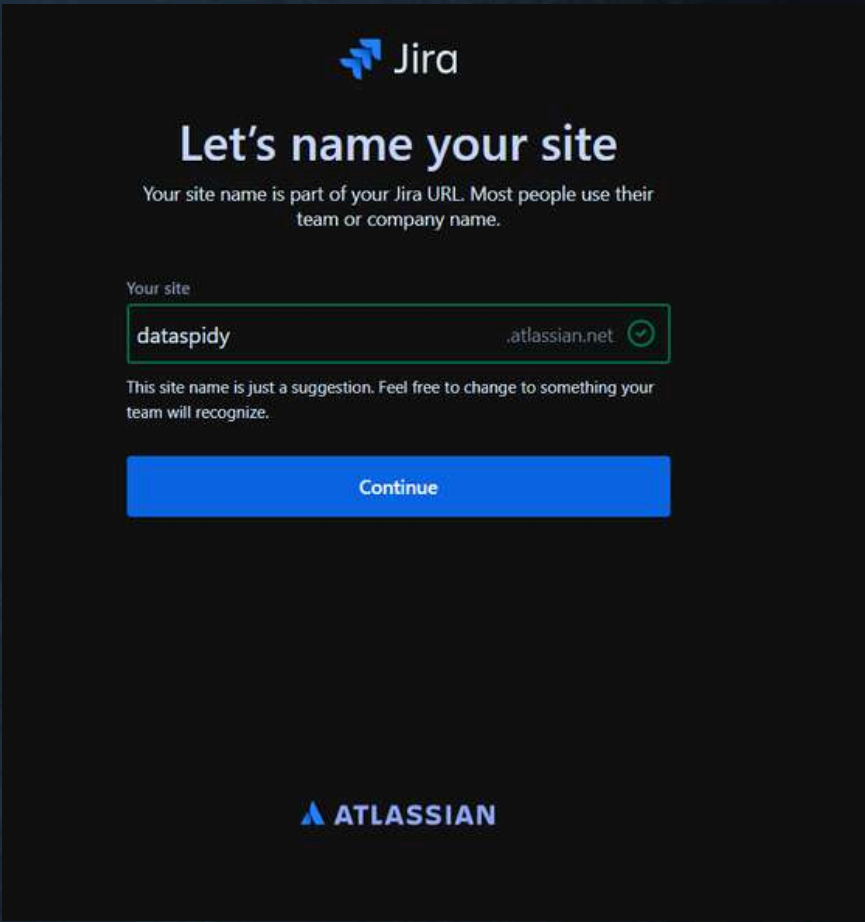
2. Create an Atlassian Account

- Step: If you don't already have an Atlassian account, you'll need to create one. Enter your email address and click on Sign up. Follow the prompts to fill in your details and create your account.



3. Set Up Your Jira Site

- Step: After creating your Atlassian account, you'll be asked to choose a name for your Jira site. This will be part of your site URL (e.g., <https://yoursitename.atlassian.net>). Enter a unique name and click on Continue.



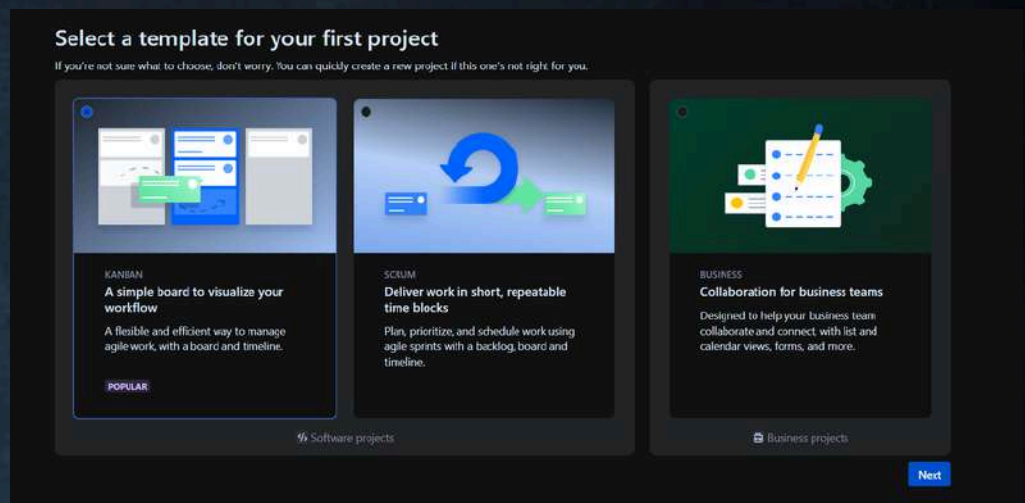
Initial Configuration

1. Choose a Template

- *Step: Jira will prompt you to choose a template based on your project type.*

For software development, you can choose between Scrum or Kanban.

Select your preferred template and click on Use template.



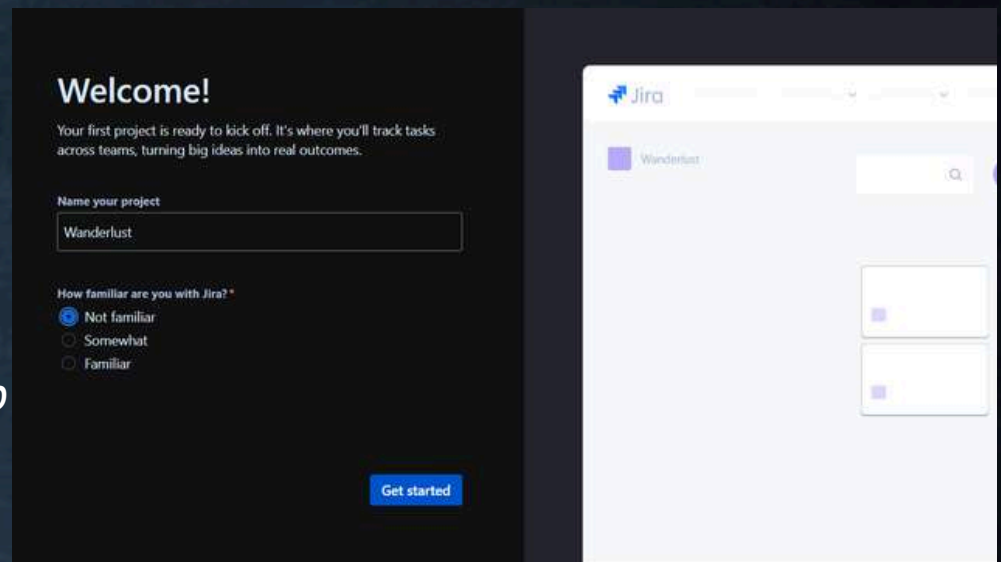
Example: Using Jira in a Project

1. *Assign Tasks: Assign these tasks to team members. For example, the designer gets the "Design Login Screen" task, and the developer gets the "Develop Login Functionality" task.*
2. *Track Progress: As team members work on their tasks, they update the status in Jira. You can see which tasks are "To Do," "In Progress," "In Review," and "Done."*
3. *Use Reports: Generate reports to see how the project is progressing. Are most tasks on track? Are there any blockers?*
4. *Integrate with Other Tools: Connect Jira with GitHub so that when the developer commits code for the login functionality, it automatically updates the task in Jira. Or integrate with Slack to get notifications about task updates.*

2. Create Your First Project

- Step: Enter a name for your project and optionally a key (a unique identifier for your project). Click on Create.

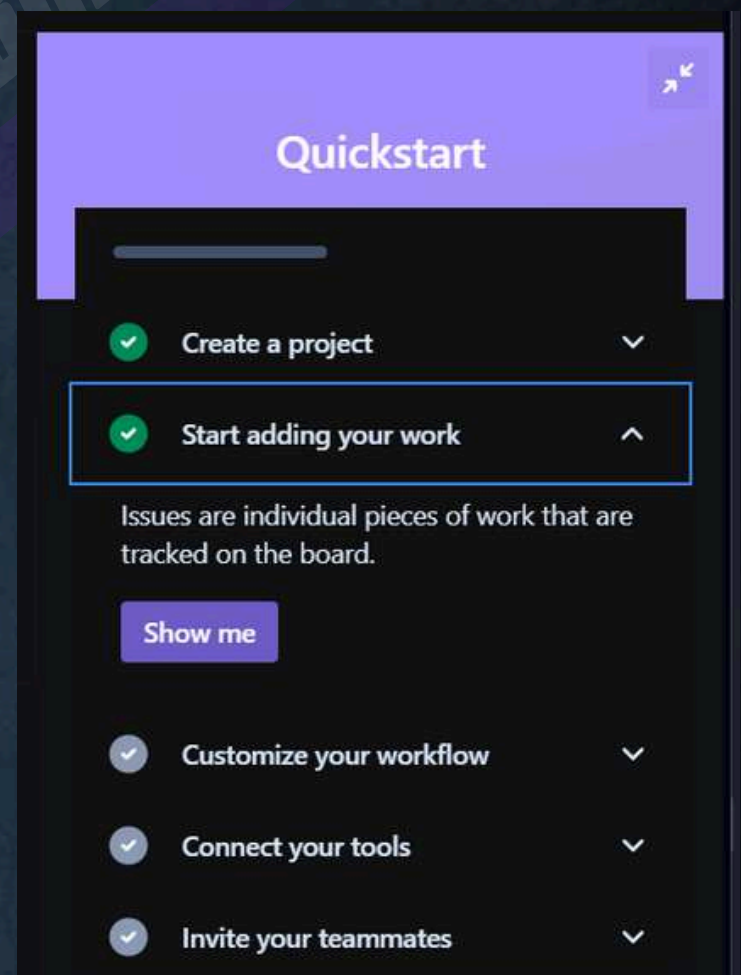
Let's say you're developing a new webapp Wanderlust. You create a project in Jira called "**Wanderlust Deployment.**"



3. Explore Jira

- Step: After setting up your project, take some time to explore the Jira interface.

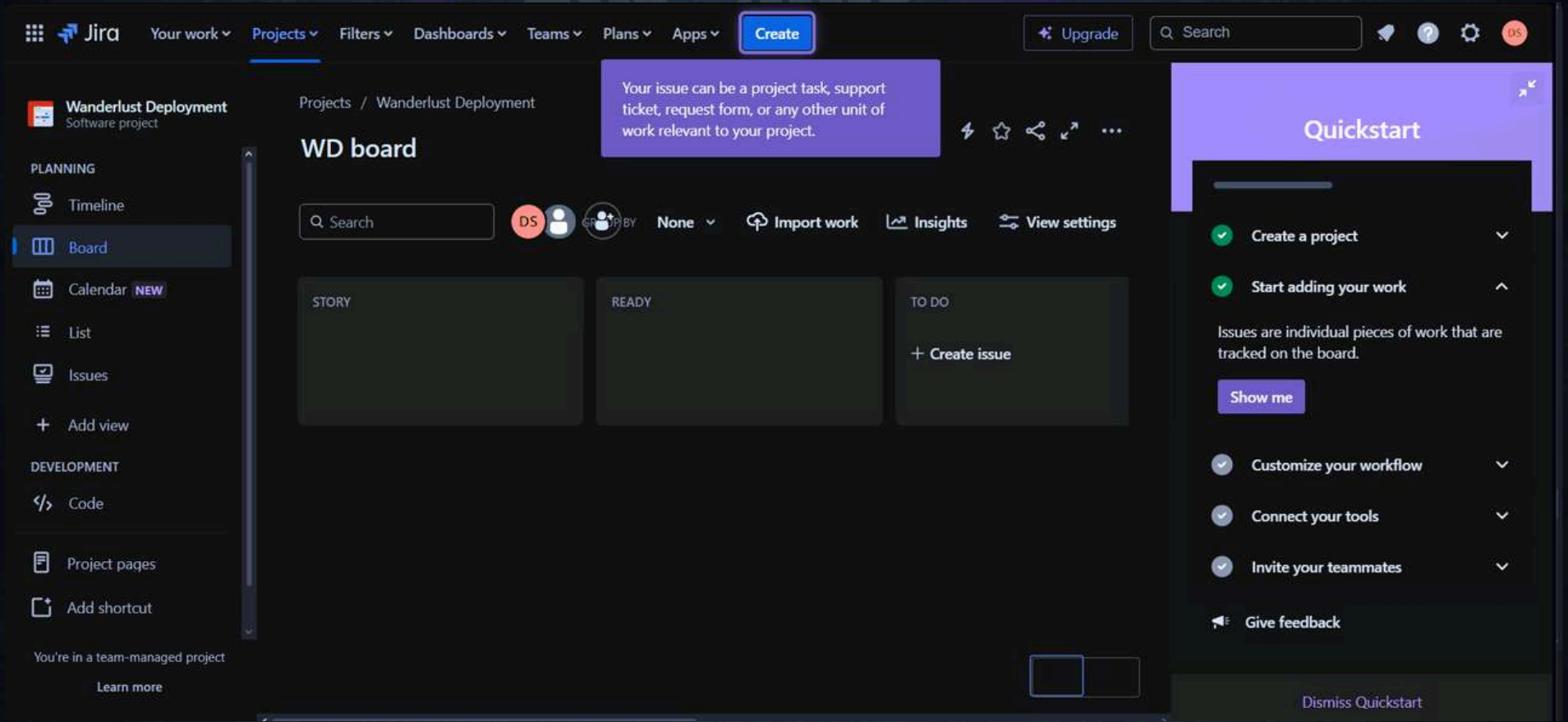
Familiarize yourself with the dashboard, backlog, boards, and other features. Also use QuickStart Panel for better step by step guidance.



4. Edit or Add Columns:

Break down the project into smaller columns, like "STORY", "READY", "TO DO", "IN PROGRESS", "REVIEW" & "DONE".

- *STORY: High-level tasks or epics.*
- *READY: Tasks that are ready to be picked up.*
- *TO DO: Tasks that need to be started.*
- *IN PROGRESS: Tasks that are currently being worked on.*
- *REVIEW: Tasks that are completed and need review.*
- *DONE: Tasks that are completed and verified.*



Add Tasks in Jira

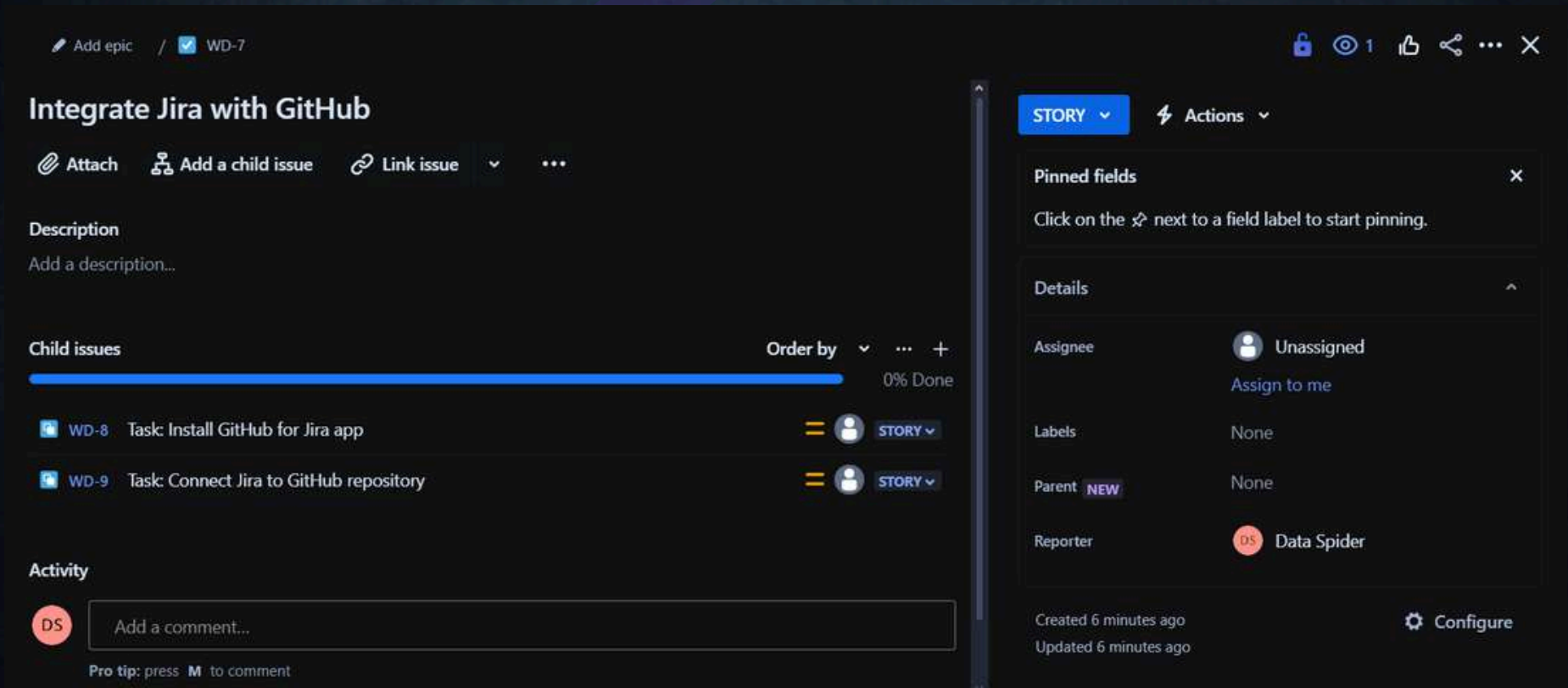
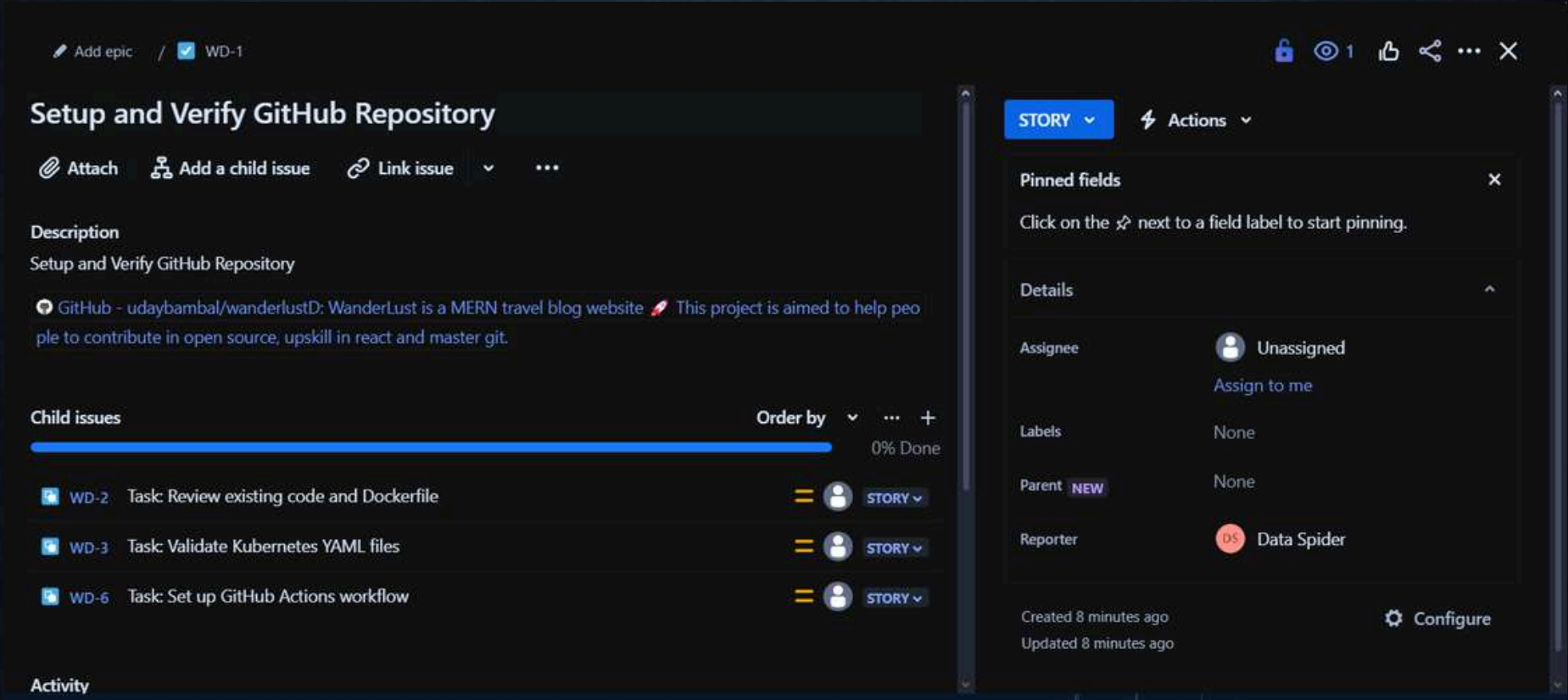
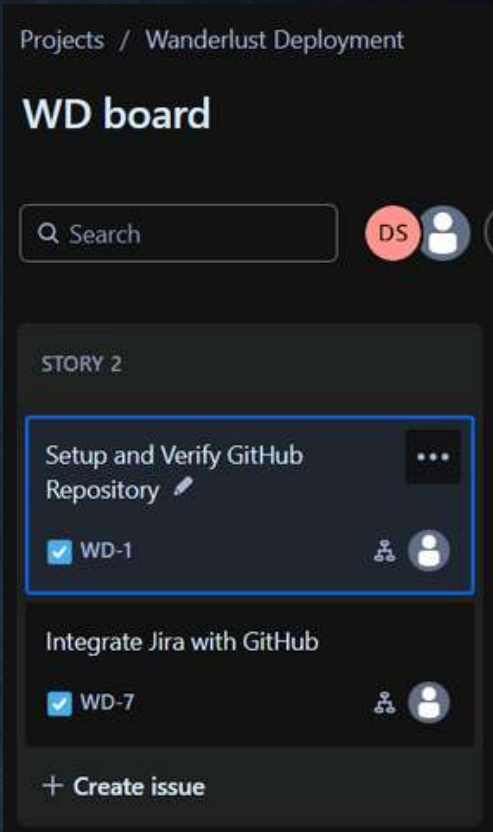
STORY

1. Setup and Verify GitHub Repository
- Task: Review existing code and Dockerfile

Task: Validate Kubernetes YAML files

Task: Set up GitHub Actions workflow
2. Integrate Jira with GitHub
- Task: Install GitHub for Jira app

Task: Connect Jira to GitHub repository



READY

Move tasks from STORY to READY once they are broken down and ready to be picked up by the team.

Task Breakdown and Workflow

TO DO

1. Review Existing Code and Dockerfile

- *Description: Ensure the Dockerfile and application code are properly set up.*
- *Assignee: [Team Member]*

Add epic

WD-10

Review Existing Code and Dockerfile

Attach

Add a child issue

Link issue

Description

Description: Ensure the Dockerfile and application code are properly set up.

Assignee: DataSpider

Activity

Show: All

Comments

History

Newest first

DS

Add a comment...

Pro tip: press **M** to comment

To Do

Actions

Pinned fields

Click on the  next to a field label to start pinning.

Details

Assignee

DS Data Spider

Labels

None

Parent

NEW None

Reporter

DS Data Spider

Created 1 minute ago

Updated 1 minute ago

Configure

2. Validate Kubernetes YAML Files

- *Description: Review deployment.yaml and service.yaml for correctness.*
- *Assignee: [Team Member]*

Add epic

WD-11

Validate Kubernetes YAML Files

Attach

Add a child issue

Link issue

Description

Description: Review deployment.yaml and service.yaml for correctness.

Assignee: DataSpider

Activity

Show: All

Comments

History

Newest first

DS

Add a comment...

Pro tip: press **M** to comment

To Do

Actions

Pinned fields

Click on the  next to a field label to start pinning.

Details

Assignee

DS Data Spider

Labels

None

Parent

NEW None

Reporter

DS Data Spider

Created 36 seconds ago

July 18, 2024 at 9:20 PM

Updated 36 seconds ago

Configure

3. Set Up GitHub Actions Workflow

- *Description: Create or verify the GitHub Actions workflow file for deployment.*
- *Assignee: [Team Member]*

Add epic

WD-12

Set Up GitHub Actions Workflow

Attach

Add a child issue

Link issue

Description

Description: Create or verify the GitHub Actions workflow file for deployment.

Assignee: DataSpider

Activity

Show: All

Comments

History

Newest first

DS

Add a comment...

Pro tip: press **M** to comment

To Do

Actions

Pinned fields

Click on the  next to a field label to start pinning.

Details

Assignee

DS Data Spider

Labels

None

Parent

NEW None

Reporter

DS Data Spider

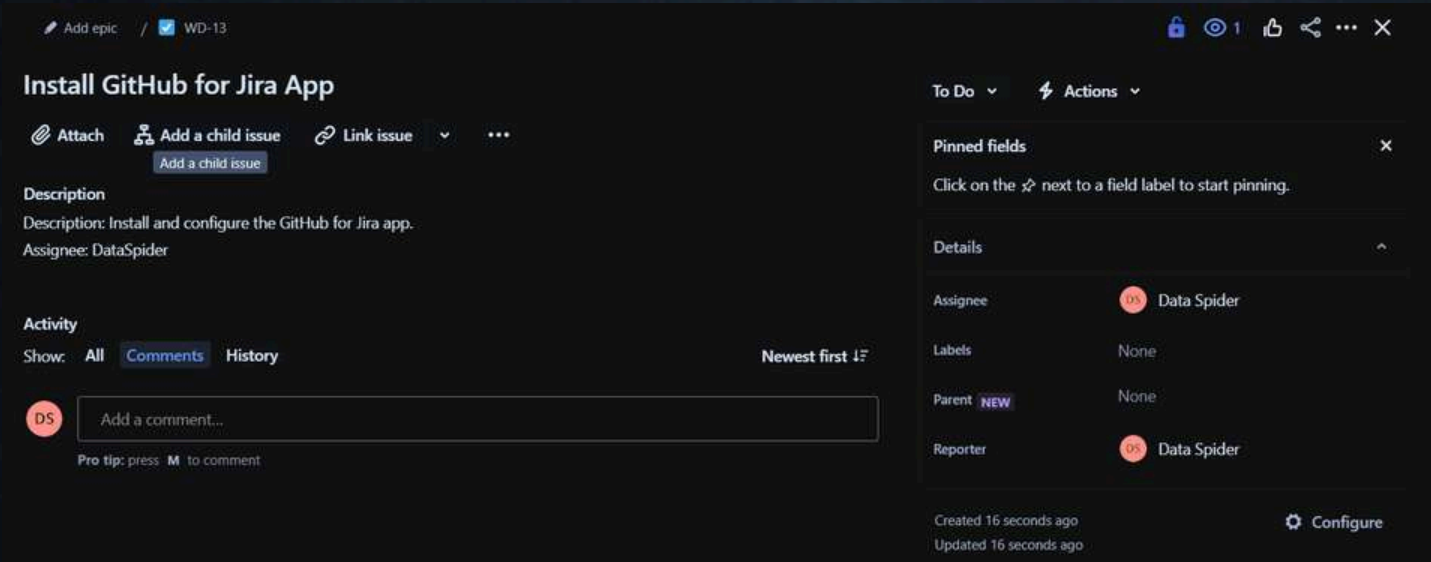
Created 17 seconds ago

Updated 17 seconds ago

Configure

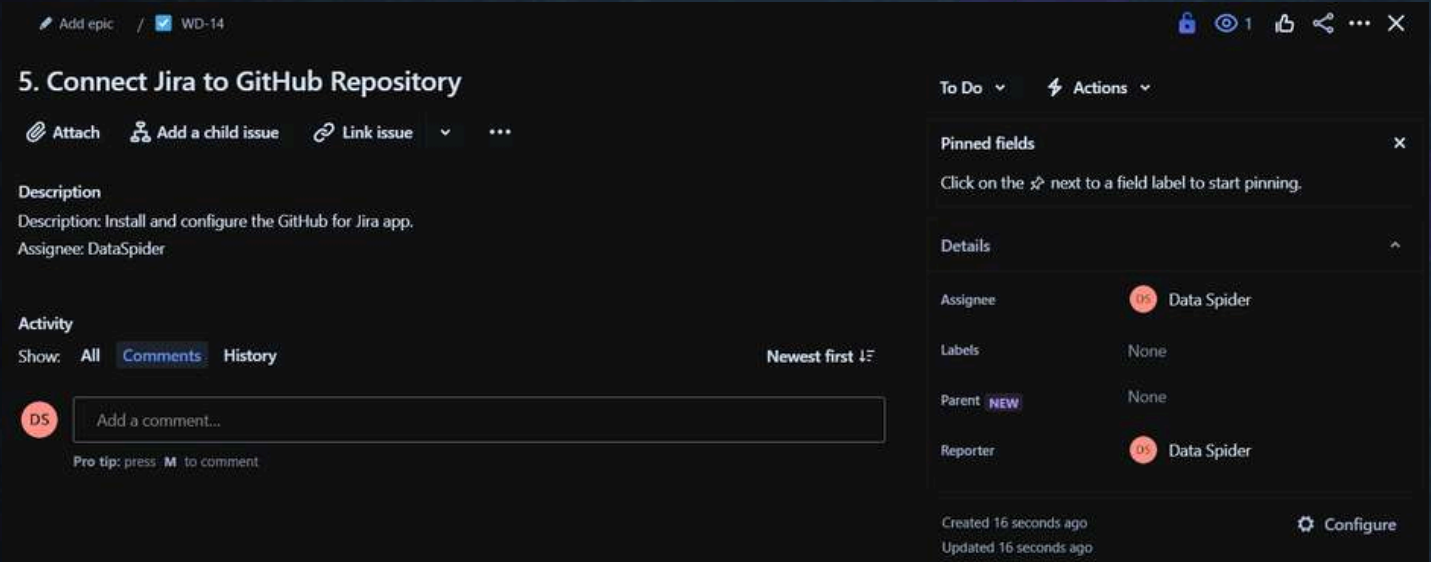
4. Install GitHub for Jira App

- Description: Install and configure the GitHub for Jira app.
- Assignee: [Team Member]



5. Connect Jira to GitHub Repository

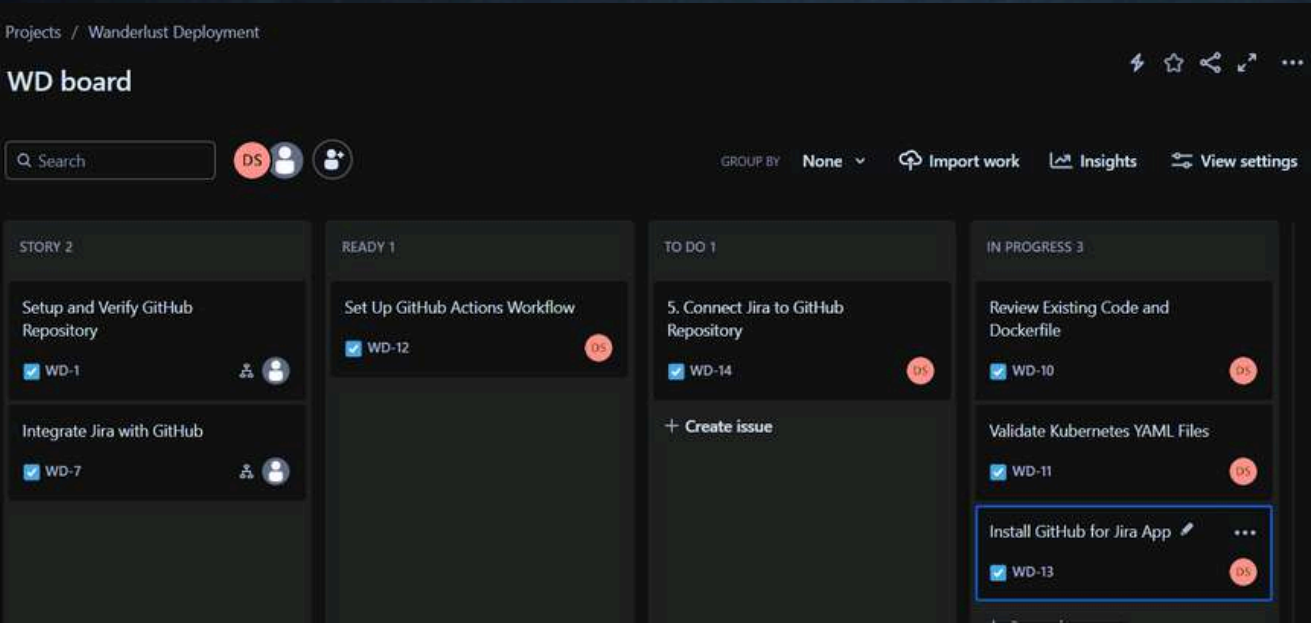
- Description: Install and configure the GitHub for Jira app.
- Assignee: [Team Member]



IN PROGRESS

Move tasks to IN PROGRESS as team members start working on them.

- 1. Review Existing Code and Dockerfile (IN PROGRESS)
- 2. Validate Kubernetes YAML Files (IN PROGRESS)
- 3. Set Up GitHub Actions Workflow (IN PROGRESS)
- 4. Install GitHub for Jira App (IN PROGRESS)
- 5. Connect Jira to GitHub Repository (IN PROGRESS)



REVIEW

Move tasks to REVIEW once they are completed and need to be reviewed.

- 1. Review Existing Code and Dockerfile (REVIEW)*
- 2. Validate Kubernetes YAML Files (REVIEW)*
- 3. Set Up GitHub Actions Workflow (REVIEW)*
- 4. Install GitHub for Jira App (REVIEW)*
- 5. Connect Jira to GitHub Repository (REVIEW)*

DONE

Move tasks to DONE once they are reviewed and verified.

- 1. Review Existing Code and Dockerfile (DONE)*
- 2. Validate Kubernetes YAML Files (DONE)*
- 3. Set Up GitHub Actions Workflow (DONE)*
- 4. Install GitHub for Jira App (DONE)*
- 5. Connect Jira to GitHub Repository (DONE)*

Detailed Steps for Each Task

Task: Review Existing Code and Dockerfile

- 1. Review the code in the GitHub repository to ensure it is correctly set up for deployment.*
- 2. Verify the Dockerfile to ensure it builds the image correctly.*
- 3. Move the task to REVIEW once completed.*

Task: Validate Kubernetes YAML Files

- 1. Check the deployment.yaml and service.yaml files in the GitHub repository for correctness.*
- 2. Ensure the configurations match the expected deployment settings.*
- 3. Move the task to REVIEW once completed.*

Task: Set Up GitHub Actions Workflow

1. Verify the existing GitHub Actions workflow file or create a new one if not present.

- Example of a workflow file (.github/workflows/deploy.yaml):

```
name: Deploy Wanderlust Web App

on:
  push:
    branches:
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v2

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v1

      - name: Login to Docker Hub
        uses: docker/login-action@v1
        with:
          username: ${{ secrets.DOCKER_USERNAME }}
          password: ${{ secrets.DOCKER_PASSWORD }}

      - name: Build and push Docker image
        run: |
          docker build -t your-dockerhub-username/wanderlust-web-app:v1 .
          docker push your-dockerhub-username/wanderlust-web-app:v1

      - name: Set up Kubernetes
        uses: azure/setup-kubectl@v1
        with:
          version: v1.19.3

      - name: Deploy to Kubernetes
        run: |
          kubectl apply -f k8s/deployment.yaml
          kubectl apply -f k8s/service.yaml
    env:
      KUBE_CONFIG_DATA: ${{ secrets.KUBE_CONFIG_DATA }}
```

Projects / Wanderlust Deployment

WD board

Q Search

DS

GROUP BY: None | Import work | Insights | View settings

STORY 2

Setup and Verify GitHub Repository

WD-1

Integrate Jira with GitHub

WD-7

READY 1

5. Connect Jira to GitHub Repository

WD-14

DS

TO DO 1

Validate Kubernetes YAML ...

WD-11

DS

+ Create issue

IN PROGRESS 2

Review Existing Code and Dockerfile

WD-10

DS

Install GitHub for Jira App

WD-13

DS

REVIEW 1

Set Up GitHub Actions Workflow

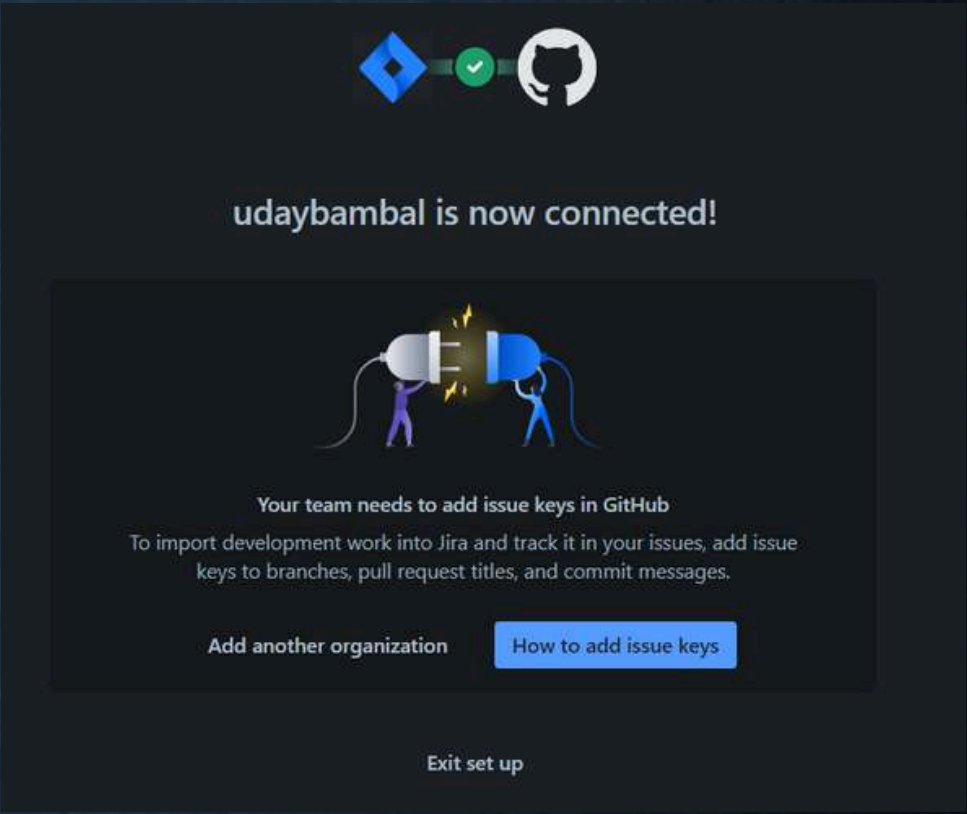
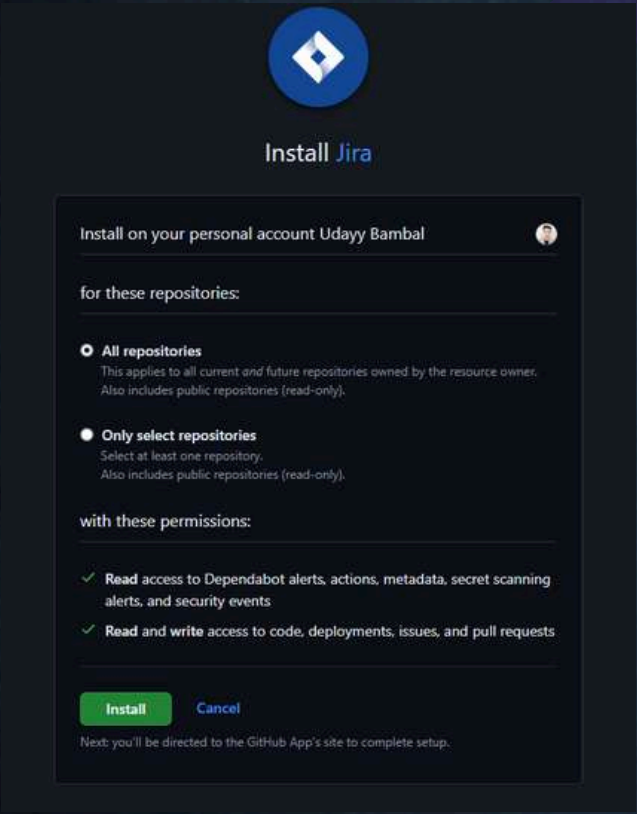
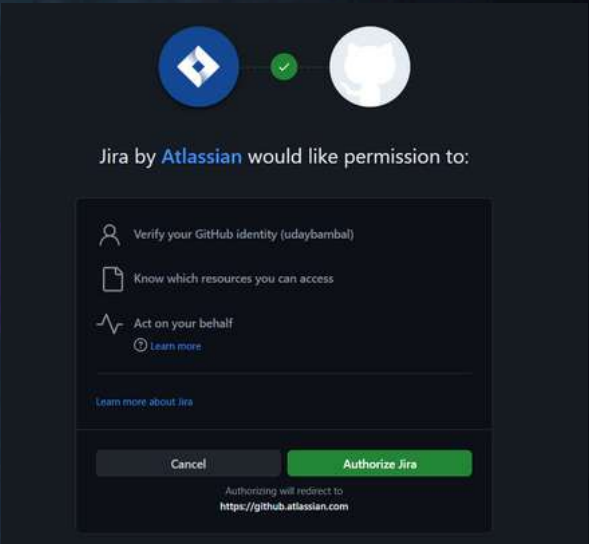
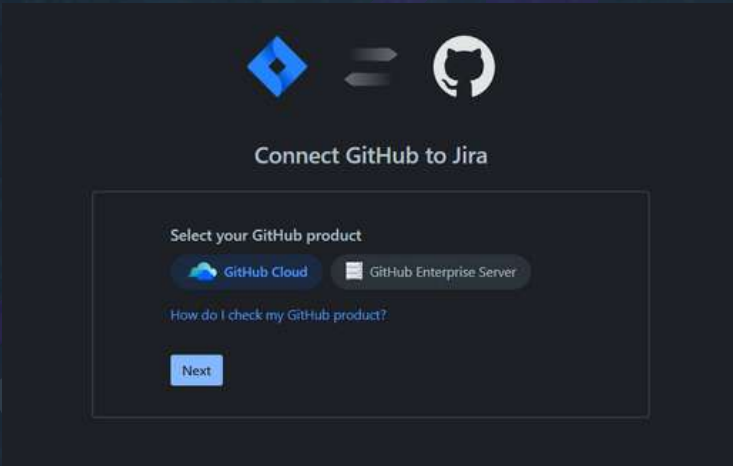
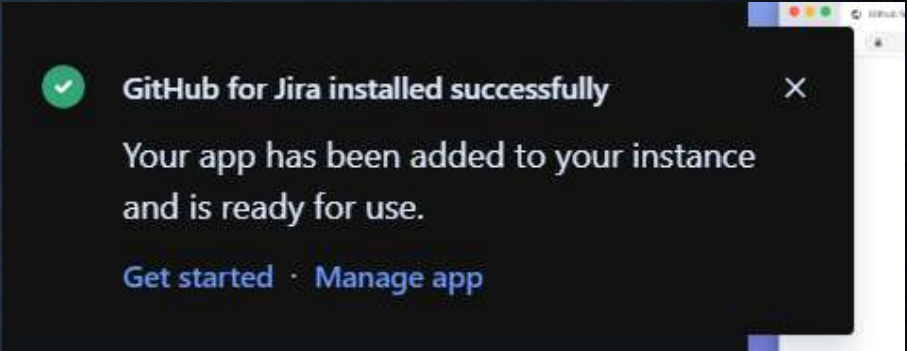
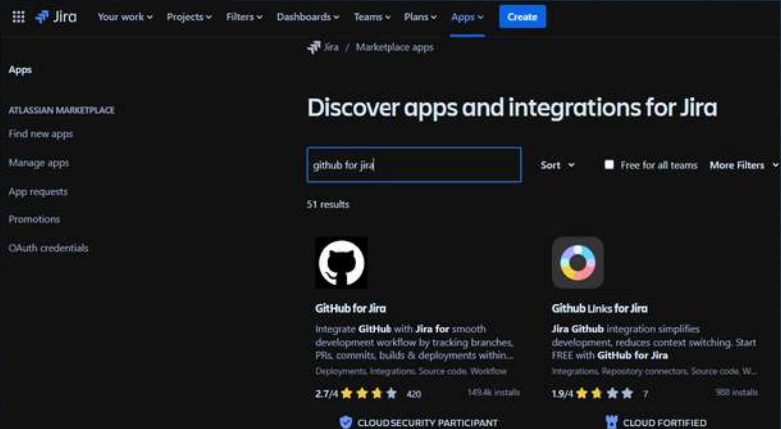
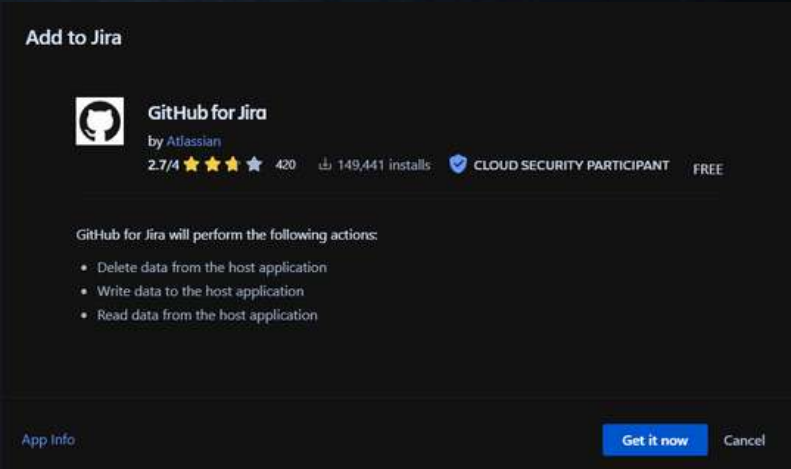
WD-12

DS

DONE

Task: Install GitHub for Jira App

- 1.Go to Jira settings.
- 2.Select Apps > Find new apps.
- 3.Search for GitHub for Jira and click Get it now.
- 4.Follow the prompts to install the app.
- 5.Move the task to REVIEW once completed.



Jira

Your work

Projects

Filters

Dashboards

Teams

Plans

Apps

Create

Upgrade

Q Search

DS

Apps

ATLASSIAN MARKETPLACE

Find new apps

Manage apps

App requests

Promotions

OAuth credentials

GITHUB FOR JIRA

Configure

GitHub configuration

Connect a GitHub organization

Connecting GitHub to Jira allows you to view development activity in the context of your Jira project and issues. To send development data from GitHub to Jira, your team must include issue keys in branch names, commit messages, and pull request titles.
Even if your organization is still backfilling historical data, you can start using issue keys in your development work immediately.

Connected organization	Repository access	Backfill status	Settings
udaybambal	Only select repos 1	FINISHED Backfilled from: 7/18/24	...

GITHUB FOR JIRA

Configure

Monitor and Manage the Project

- *Daily Standups: Regularly update the status of tasks.*
- *Review Sessions: Regularly review tasks moved to the REVIEW column.*
- *Retrospectives: After project completion, review what went well and what could be improved.*

Jira

Your work

Projects

Filters

Dashboards

Teams

Plans

Apps

Create

Upgrade

Q Search

DS

Wanderlust Deployment

Software project

PLANNING

Timeline

Board

Calendar

List

Issues

Add view

DEVELOPMENT

Code

Project pages

Add shortcut

Project settings

Projects / Wanderlust Deployment

WD board

Q Search

DS MP DS

GROUP BY: None Import work Insights View settings

STORY 2	READY 1	TO DO 1	IN PROGRESS 1	REVIEW 1	DONE 1
Setup and Verify GitHub Repository WD-1	5. Connect Jira to GitHub Repository WD-14	Validate Kubernetes YAML Files WD-11 Create issue	Review Existing Code and Dockerfile WD-10	Set Up GitHub Actions Workflow WD-12	Install GitHub for Jira App WD-13

By following this detailed plan, you ensure a structured approach to managing the deployment of the Wanderlust web app using Kubernetes, integrated with Jira and GitHub for effective project management and continuous deployment.